

# Software Testing

University of Oslo, **Department of informatics**

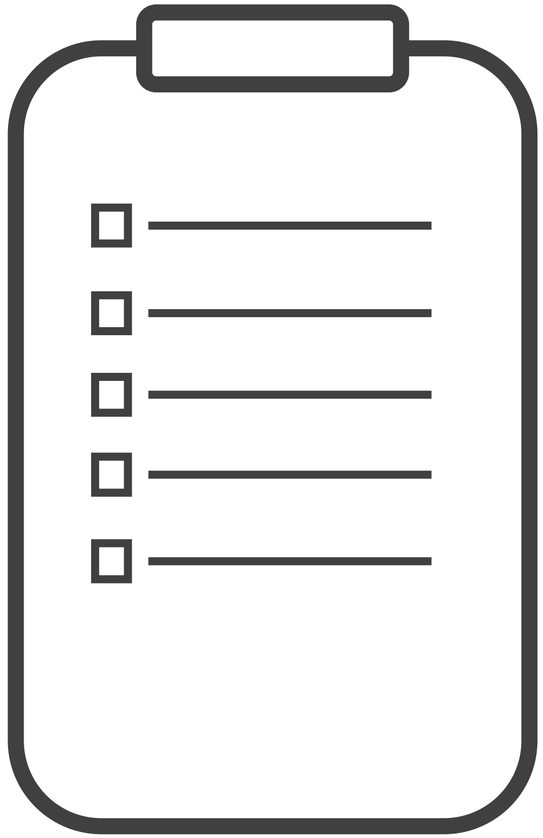
# Agenda

Introduction

Course Structure

Recap of the first week

Weekly exercises



# Summary

Why is testing necessary?

What is testing?

Test principles

- Seven test principles in total

Fundamental test process

- Five important activities in total

Psychology of testing

# Part I: Close-ended questions

# Question 1

A company recently purchased a commercial off-the-shelf application to automate their bill-paying process. They now plan to run an acceptance test against the package prior to putting it into production. Which is their most likely reason for testing?

a. To build confidence in the application
b. To detect bugs in the application
c. To gather evidence for a lawsuit
d. To train the users

# Question 1: Clues

Which is their most likely reason for testing?

- Characteristics of acceptance testing
  - Establish confidence the system | part of system
  - Examine non-functional characteristics of the system
  - Verify the fitness for use
  - Responsibility lies with the customers / users
  - Different types of acceptance testing
    - User acceptance testing
    - Operational testing
    - Contract and regulation acceptance testing

# Question 2

According to the ISTQB Glossary, the word ‘bug’ is synonymous with which of the following words?

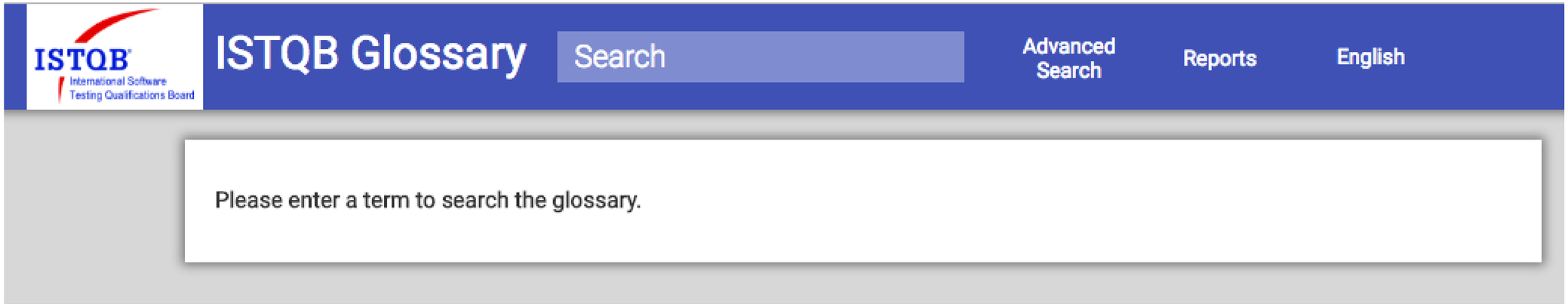
a. Incident
b. Defect
c. Mistake
d. Error

# Question 2: Clues

According to the ISTQB Glossary, the word ‘bug’ is synonymous with which of the following words?

ISTQB Glossary

https://glossary.istqb.org/



# Question 2: Clues

According to the ISTQB Glossary, the word ‘bug’ is synonymous with which of the following words?

Incident

*“Any event occurring that requires investigation”*

Defect

*“Flaw in a component or system that can cause the component or system to fail to perform its required function, e.g. an incorrect statement or data definition.“*

Mistake / Error

*“A human action that produces an incorrect result”*

# Question 3

According to the ISTQB Glossary, a _____ relates to negative consequences that could occur.

# Question 3: Clues

According to the ISTQB Glossary, a _____ relates to negative consequences that could occur.

What could go wrong?

Estimation and scheduling

Sudden changes to requirements

Employees leaving, causing delays

Developers showing off skills, adding unnecessary features

These factors could result in *"future negative consequences"*

How to handle them? → Risk management

# Question 4

Ensuring that test design starts during the requirements definition phase is important to enable which of the following test objectives?

a. Preventing defects in the system
b. Finding defects through dynamic testing
c. Gaining confidence in the system
d. Finishing the project on time

# Question 4: Clues

Why is it important to start test design during the requirements definition phase?

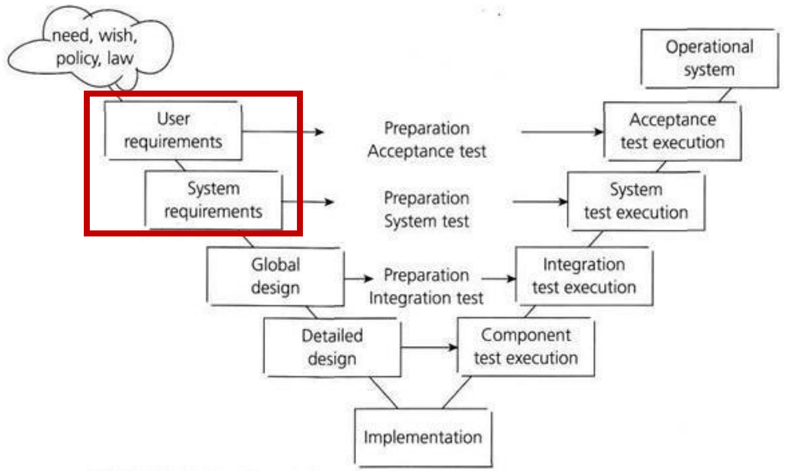
What is the purpose of testing?

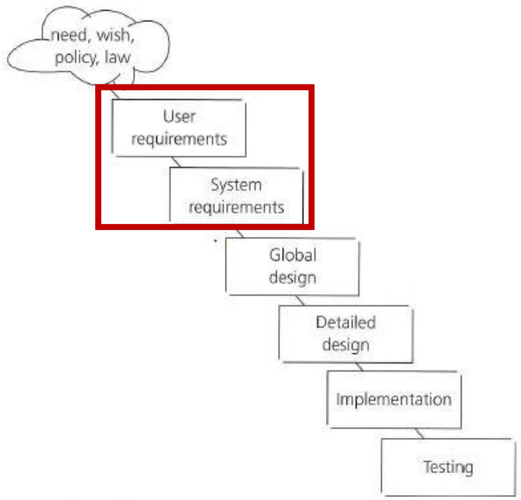
- Finding defects
  - Reduce probability of undiscovered defects
- Preventing defects
  - Ensure, as far as possible, that defects are not introduced
- Gaining confidence in the level of quality
  - Presence / Absence of defects to indicate SW system quality
- Provide information for decisions-making
  - Testing grants different *insight* into various aspects of software

# Question 4: Clues

Why is it important to start test design during the requirements definition phase?

Why start during the requirements definition phase?




Early testing

Testing should start as soon as possible

# Question 5

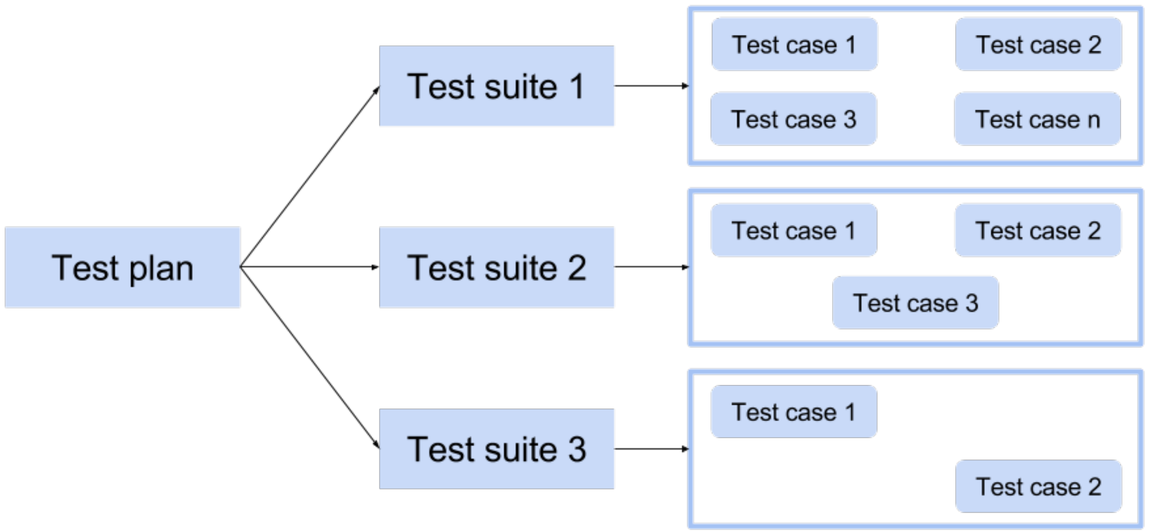
An exhaustive test suite would include:

a. All combinations of input values and preconditions
b. All combinations of input values and output values
c. All pairs of input values and preconditions
d. All states and state transitions

# Question 5: Clues

An exhaustive test suite would include:

- Test suite
  - Logical collection of several test cases
  - Test software program to show specified set of behaviours



# Question 6

According to the ISTQB Glossary, regression testing is required for what purpose?

a. To verify the success of corrective actions
b. To prevent a task from being incorrectly considered completed
c. To ensure that defects have not been introduced by a modification
d. To motivate better unit testing by the programmers

# Question 6: Clues

According to the ISTQB Glossary, regression testing is required for what purpose?

Regression testing

After fixing a defect; have we introduced new defects?

Checks unchanged areas of the software

Confirmation testing

Retesting of software after defect has been detected and fixed

Confirm that the original defect has been successfully removed

# Question 7

Which of the following is most important to promote and maintain good relationships between testers and developers?

a. Understanding what managers value about testing
b. Explaining test results in a neutral fashion
c. Identifying potential customer work-arounds for bugs
d. Promoting better quality software whenever possible

# Question 7: Clues

Which of the following is most important to promote and maintain good relationships between testers and developers?

- Importance of good team dynamics
  - People execute the process → People affect the result
  - Helping developers improve → Better relationship to testers
- Promote and nurture trusting inter-team relationships
  - Constructive feedback
  - Learning opportunities

# Question 8

Which of the statements below is the best assessment of how test principles apply across the test life-cycle?

a. Test principles only affect the preparation for testing
b. Test principles only affect test execution activities
c. Test principles affect the early test activities such as review
d. Test principles affect activities throughout the test life-cycle

# Question 8: Clues

Which of the statements below is the best assessment of how test principles apply across the test life-cycle?

**P1: Testing shows presence of defects**
- Testing can show that defects are present, but cannot prove there are no defects.
- Testing reduces the probability of undiscovered defects remaining in the software; but even if no defects are found, this is not a proof of correctness.

**P2: Exhaustive testing is impossible**
- Testing everything is not feasible. We use risks and priorities to focus on test effort.

**P3: Early testing**
- Testing should start as soon as possible in the development life-cycle and should be focused on defined objectives.

**P4: Defect clustering**
- A small number of modules contain most of the defects discovered during pre-release testing.

**P5: Pesticide paradox**
- If the same set of tests will be repeated over and over, it will no longer find new bugs.

**P6: Testing is context dependent**
- I.e. safety-critical SW is tested differently from an e-commerce site.

**P7: Absence-of-errors fallacy**
- Finding and fixing defects does not help if the SW system is unstable or does not meet the user's expectations.

# Question 8: Clues

Which of the statements below is the best assessment of how test principles apply across the test life-cycle?

- Test principles deal with
  - Good practices for software testing
    - Early testing
  - Importance of testing throughout the test life-cycle
  - Limitations of testing
    - Exhaustive testing / Presence of defects
  - Considerations and challenges of software testing

# Question 9

Is “evaluation of exit criteria” for of the fundamental test process?

a. True
b. False

# Question 9: Clues

Is “evaluation of exit criteria” for of the fundamental test process?

Fundamental test process

Testing is *not* a single activity

Activities for each step in the test process → From planning to test closure

Activities

1 Plan and Control
2 Analysis and Design
3 Implementation and Execution
4 Evaluate Exit Criteria and Report
5 Test Closure Activities

# Question 9: Clues

Is “evaluation of exit criteria” for of the fundamental test process?

1. Plan and Control

   What / How / When / By whom?

2. Analysis and Design

   Review test basis / Analyse test objectives

   Design test cases, environments, data

3. Implementation and Execution

   Group, prioritise test scripts / Write automated test scenarios

   Run tests / Compare results / Repeat / Log outcome

# Question 9: Clues

Is “evaluation of exit criteria” for of the fundamental test process?

4. Evaluate Exit Criteria and Report

   Assess test execution against defined objectives

   Definition of Done (DoD)

   Write / Extract test summary / Produce report for stakeholders

5. Test Closure Activities

   Assess test effort: What has been delivered?

   Archive delivered items (SW, tests, documentation) / Analyse lessons learned

# Question 10

Pair the following processes with their corresponding activities:

| | |
|---|---|
| 1. Test plan | A. Group tests into scripts |
| 2. Test analysis | B. Write or extract a test summary report for the stakeholders |
| 3. Test implementation | C. Transform test objectives into test conditions and test cases |
| 4. Test reporting | D. Establish the scope, objectives, and risks of testing |

# Part II: Exercises and Open-ended questions

# Exercise

A test team consistently finds between 90 and 95% of the defects present in the system under test. While the test manager understands that this is a good defect-detection percentage, senior management and executives remain disappointed, saying that the test team misses too many bugs. Given that the users are generally happy with the system and that the failures which have occurred have generally been low impact, which of the testing principles is most likely to help the test manager explain why some defects are likely to be missed?

Exhaustive testing is impossible
Defect clustering
Pesticide paradox
Absence-of-errors fallacy

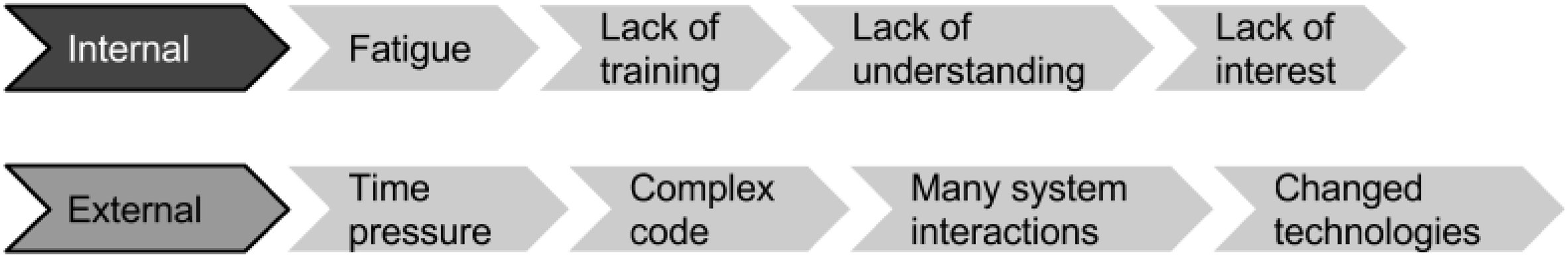
# Open-Ended Questions

Can you give examples of internal and external causes of defects? Which one do you consider to be the most important?

Why is it important to check a software system if it is fit for use?

What qualities should a tester have, if you were to hire him?

# Internal and External Causes



# Checking if SW is Fit for Use

Fit for use → Acceptance testing

- Does the software product meet the customer's needs?
- Does the software product comply with the specified requirements / documentation?
- Last chance of the customer to discard an insufficient / inadequate product
- Adequate testing protects customer from losses caused by poor software products
- Compares results to expectations

Validation

- Did we build the *right thing*?

Verification

- Did we build the *thing right*?

# Qualities of a Tester

Problem-oriented

- Desire to find problems / Ability to solve problems

Detail-oriented

Curiosity and Creativity

- Approaching the test effort from new, varied angles

Professional pessimism

- Being realistic about the product / human behavioural patterns

Good interpersonal and communication skills

- Courtesy, understanding of others, good attitude, diplomatic, etc.

# The seminar slides are made by

Yulai Fjeld, revised by Eva H. Vihovde